



Week 6 Lecture 6

▼ Class	BSCCS2003
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🔗 Materials	
# Module #	37
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Important Concepts of an API

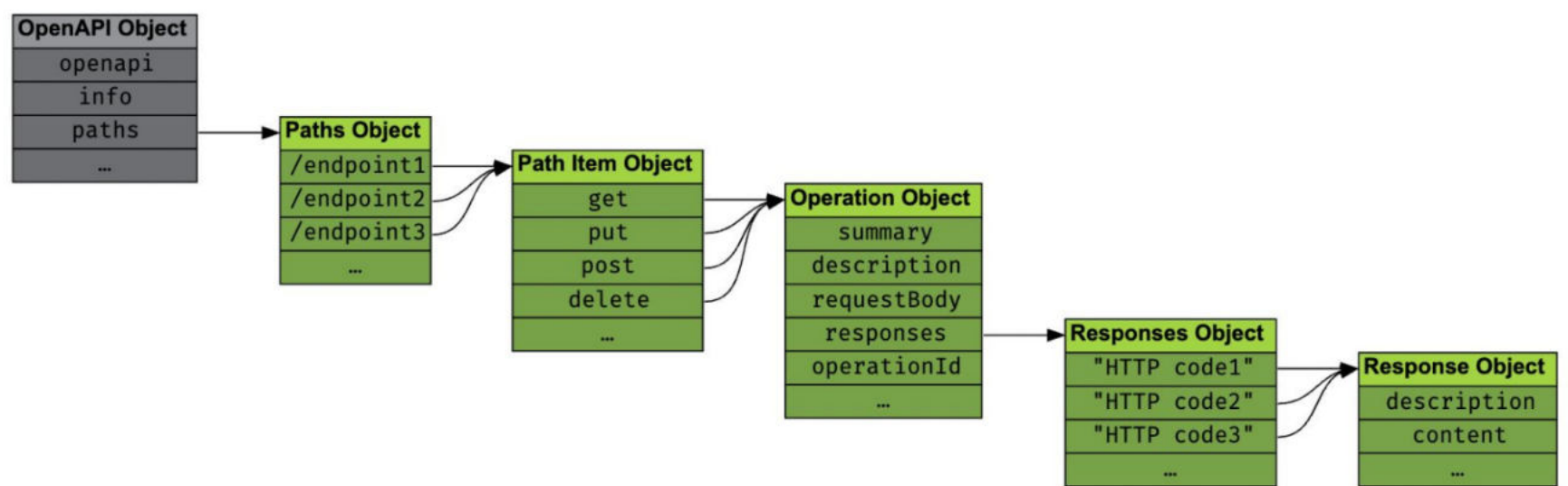
Concepts

- Describe in YAML (or possibly JSON)
- Specific structure to indicate overall information, paths, schemas, etc

eg:

```
openapi: 3.1.0
info:
  title: A minimal OpenAPI document
  version: 0.0.1
paths: {} # No endpoints defined
```

Endpoints List



Source: <https://oai.github.io/Documentation/specification-paths.html>

Paths

```
openapi: 3.1.0
info:
  title: Tic Tac Toe
  description: |
    This API allows writing down marks on a Tic Tac Toe board
    and requesting the state of the board or of individual squares
  version: 1.0.0
paths:
  /board:
  ...
```

Operations

```
paths:
  /board:
    get:
      ...
    put:
      ...
```

Operation object

```
paths:
  /board:
    get:
      summary: Get the whole board
      description: Retrieves the current state of the board and the winner.
      parameters:
        ...
      responses:
        ...
```

Responses

```
paths:
  /board:
    get:
      responses:
        "200":
          ...
        "404":
          ...
```

Response Objects

```
paths:
  /board:
    get:
      responses:
        "200":
          description: Everything went fine.
```

```
content:
  ...
```

Content Specification

```
content:
  application/json:
    ...
  text/html:
    ...
  text/*:
    ...
```

Schema

```
content:
  application/json:
    schema:
      type: integer
      minimum: 1
      maximum: 100
```

Complex schema

```
content:
  application/json:
    schema:
      type: object
      properties:
        productName:
          type: string
        productPrice:
          type: number
```

Parameters

```
paths:
  /users/{id}:
    get:
      parameters:
        - name: id
          in: path
          required: true
```

Request body

```
requestBody:
  content:
    application/json:
      schema:
        type: integer
        minimum: 1
        maximum: 100
```

Best practices

- Design-first v/s Code-first
 - Always prefer the design-first
- Single source of truth
 - The structure of the code should be derived from the OAS - or -
 - Spec should be derived from the code
 - Minimize chances of code and documentation diverging
- Source code version control
- OpenAPI is ... Open — public documentation better to identify the problems

- Automated tools, editors — make use of them